

A new lattice-based authentication scheme for IoT[☆]

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ARTICLE INFO

Keywords:

Radio-frequency identification tags
Internet of things
Lattice-based
Authentication scheme
Post-quantum cryptography

ABSTRACT

In this paper, we propose a lattice-based authentication scheme for IoT technologies such as RFID systems. The security of the proposed scheme is based on ISIS problem which helps to resist to quantum attacks. The proposed scheme satisfies zero-knowledge property. Moreover, the location privacy, unforgeability, scalability and forward secrecy properties are achieved. The proposed scheme also has a resistance to man in the middle, replay and impersonation attacks. Moreover, its efficiency is comparable with the recent studies.

1. Introduction

The concept of the Internet of Things (IoT) has been developing very rapidly over past few years. IoT can be defined as a system with devices that can transfer data over their network with interconnected computing devices, mechanical and digital machines, objects or unique identification numbers [1]. The concept and development of IoT are examined while the challenges of its development, the expected future growth and open problems are discussed in [2]. The fact that IoT is used almost every area of our lives, brings along the security problems. In [1], the security analysis of IoT architecture, the classification of security challenges and defense mechanisms are examined. In addition, the solutions are given against IoT attacks.

In IoT systems, the authentication schemes integrated in RFID tag are used to secure the device [3]. RFID systems have three parts, the RFID tag, the reader and the server, respectively. In an RFID system, the tag, which has a unique ID, is matched to an object. Owing to this tag, the reader can identify the object. In case of the reader requires any related information about the tag, the reader can get the information from the server. The communication between the tag and the reader is supposed to be vulnerable to attacks and is called insecure, on the contrary, the channel between the reader and the server is called secure.

Several authentication schemes have been put forward to provide security and properties which are required for RFID applications. Many authentication schemes have been proposed for RFID applications [4–9]. In [4], Chien and Huang developed a lightweight authentication protocol, based on only bitwise XOR, secret information SID and pseudo random numbers, for low-cost RFIDs. Their scheme provides no-traceability, forward secrecy, mutual authentication while it resists to

replay attack and DoS attack. Huang and Jiang proposed two improved authentication schemes by using ultralightweight operations such as PRNG, XOR and the cyclic redundancy check (CRC) [8]. These schemes resist the forged-tag, forged-server, man in the middle (MITM), replay, tracking, forward secrecy and DoS attacks. In [5], a hash-based RFID tag mutual authentication scheme was constructed. The attackers need a high cost to achieve brute-force attack. Moreover, the scheme satisfies mutual authentication, confidentiality, indistinguishability and forward secrecy. In the scheme of Dass and Om [6], pseudorandom number generators (PRNG), hash functions and exclusive-or (XOR) operator were used for authentication. For this scheme, the security analysis for different attacks on RFID and the performance analysis are given. Dong et al. discussed a cloud-based lightweight RFID scheme [7]. In this scheme, the authentication process is achieved with hash function. While it ensures mutual authentication between the reader and the tag, it also performs the security of communication between the reader and the cloud database. In addition, it is shown that the proposed scheme is secure against replay, desynchronization and DoS attacks. In [9], Wu proposed a matrix-based authentication scheme. The scheme, in which Hill cipher hard problem was used, satisfies confidentiality, unforgeability, location privacy and scalability. On the other hand, the concept of the authentication is not bounded only with RFID. Kumar et al. proposed an authentication scheme based elliptic curve cryptography (ECC) for wearable devices in [10]. The formal analysis was given using the random oracle model. The scheme is resistant to password guessing attack, wearable device stolen attack, man-in-the-middle attack, replay attack, denial-of-service attack and impersonation attack. In [11], Kumari et al. constructed an ECC based authentication scheme for smart card. For many security attacks and properties, the

[☆] This research was partially supported by TUBITAK under grant no. 117E636.

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detailed security analysis was given. The formal security analysis of the scheme was made by depending on random oracle model. The scheme satisfies many security properties like smart card loss attack, anonymity, mutual authentication, replay attack, impersonation attack, server spoofing attack, parallel attack, forward secrecy etc. Note that the number of authentication schemes which are secure against post-quantum attacks is limited.

Quantum computers will significantly increase the computation power. In addition, Shor proposed an algorithm which solve hard problems such as discrete logarithm and factorization problem in quantum computers in polynomial time [12]. In the report published by National Institute of Standards and Technology (NIST), public key cryptosystems such as RSA, DSA and ECDSA which are believed secure for traditional computation systems, will become insecure for post-quantum [13]. Therefore, new authentication schemes need to be constructed for post-quantum RFID applications. It is believed that systems that are based on hash, code, lattice, multivariate polynomials and isogeny are secure since there is no proposed polynomial-time algorithm to solve these problems in quantum computers [14]. In [15], Silva et al. surveyed lattice-based identification schemes. They gave some information such as interactive proof properties, cost, memory requirements and security for each scheme. It is known that it is important to construct identification and authentication schemes for post-quantum era. In this paper, we propose a lattice-based authentication scheme.

1.1. Motivation and our contribution

Nowadays, RFID systems in a variety of areas have been utilized. Many authentication schemes are proposed for RFID systems. The concept of security and privacy has gained importance with the spread of RFID in every field. Efficient schemes providing all security requirements must be created since the attackers have more computational power day by day.

In this paper, we discuss an identification scheme given in [16]. We propose a lattice-based authentication scheme for RFID systems by using the scheme in [16]. In the proposed scheme, the authentication process is carried out the basis of ISIS problem. Note that the lattice-based problems cannot be solved in polynomial-time even in quantum computers. Therefore, the proposed authentication scheme is secure against to quantum attacks unlike the existing schemes for RFID devices. In addition, the proposed scheme is zero-knowledge. This property ensures that any attacker cannot obtain information about the scheme during the communication. We present an authentication scheme providing all security requirements needed for RFID systems.

1.2. Organization

The rest of this paper is organized as follows. In Section 2, we firstly give some definitions. Then, we present a new authentication scheme based on ISIS problem. In Section 3, we give security analysis for the proposed scheme. In addition, we compare the proposed scheme with the previous ones. In Section 5, the conclusion and future works are stated.

2. The proposed authentication scheme

In this section, we recall computationally hard problem ISIS in lattices. Then, we transform an identification scheme based on the hardness of ISIS problem [16] into a 3-pass authentication scheme for IoT. The proposed scheme uses the benefits of lattice-based cryptography. Therefore, we obtain a post-quantum secure authentication scheme. In Definition 1, ISIS problem is given.

Table 1
Notations.

Symbol	Definition
x	The identification number of the tag
u, P_σ	The secret keys of the tag
x_1	The random value generated by the tag
A, y	Shared keys between the tag and the server
P_σ^{-1}	The secret key of the server (The inverse of P_σ)
H_1, H_2	One-way hash functions
α	The random value generated by the reader

Definition 1 (Inhomogeneous Small Integer Solution (ISIS) Problem: [17]). Given a modulus q , a matrix $A \in \mathbb{Z}_q^{n \times m}$ whose columns are m uniformly random vectors $a_i \in \mathbb{Z}_q^n$, a vector $y \in \mathbb{Z}_q^n$, find an integer vector $x \in \mathbb{Z}_q^m$ such that:

$$Ax = \sum_i a_i x_i = y \in \mathbb{Z}_q^n$$

Note that a special case of the ISIS problem is SIS problem in which y equals zero. We propose an authentication scheme using ISIS problem. The notations, used to construct the proposed scheme, are defined in Table 1.

The proposed scheme given in Fig. 1 consists of the following two phases.

2.1. Setup phase

The hash functions are embedded in each RFID tag in terms of cost. The server uses also the same hash functions. Firstly, the server generates P_σ , A and y for each tag. Each tag has a unique identification number denoted with x in the proposed scheme. The result is a non-zero solution vector of ISIS problem which satisfies the condition $Ax = y \pmod{q}$. Each tag's identification number x , P_σ^{-1} and y are stored in the server's database. The server also computes the result of the multiplication AP_σ^{-1} to use the verification of the commitment values. The multiplication AP_σ^{-1} , known by the server, is a precomputed value for each tag. In addition, A and y are shared keys between the tag and the server. Before the authentication phase begins, the tag chooses randomly the secret key $u \in \mathbb{Z}_q^m$, having the related properties to satisfy that ISIS problem is computationally hard. In addition, the tag randomly chooses $x_1 \xleftarrow{\$} \{0, 1\}^m$. Note that all computation in the proposed scheme is performed modulo q and the following notations are used. S_m is the set of all permutations on m elements. Any permutation $\sigma \in S_m$ is a linear operation, P_σ is the associated $m \times m$ binary matrix and A is a matrix chosen in $\mathbb{Z}_q^{n \times m}$ by providing $y = Ax \pmod{q}$.

2.2. Authentication phase

After choosing the unique id x , x_1, u, q, P_σ and A , the authentication phase starts. This phase performs interactively among the tag, the reader and the server.

Step 1: The reader chooses a random value $\alpha \in \mathbb{Z}_q$ and sends it to the tag with a query signal.

Step 2: In this step, we have two steps:

Step 2.1: The tag receives the random value α and computes four values $\beta = P_\sigma(u + \alpha x)$, $c_1 = P_\sigma(x + x_1)$, $c_2 = H_1(c_1, Au)$, $c_3 = Au + x$. The tag sends four commitment values to the reader.

Step 2.2: The second phase is carried out between the reader and the server. Once the reader takes the commitment values, the reader forwards the received values β, c_1, c_2, c_3 with its random value α to the server.

Step 3: The server computes $c_4 = AP_\sigma^{-1}\beta - y\alpha$ and $c_5 = c_3 - c_4$ to obtain the id x . Then, the server checks whether c_2 equals $H_1(c_1, c_4)$ or not. If the equation is verified, the server realizes that the server communicate with the valid tag. Otherwise, the authentication process

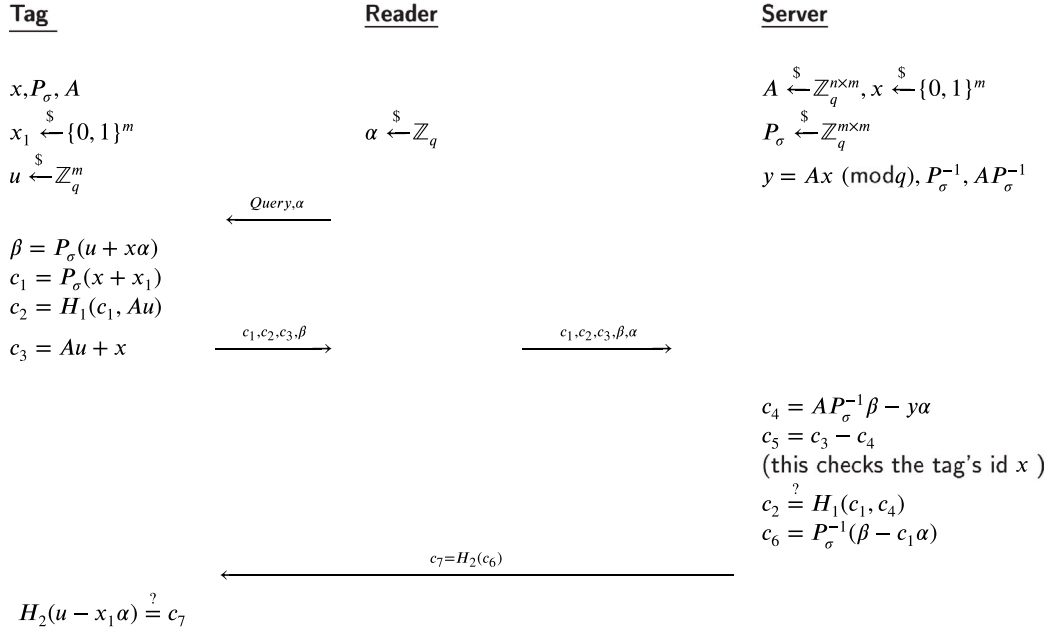


Fig. 1. The proposed authentication scheme.

is terminated by informing the reader. Finally, the server generates $c_6 = P_\sigma^{-1}(\beta - c_1\alpha)$ and sends c_6 to the tag after hashing it. The tag receives c_7 and checks whether the following equation is true or not.

$$H_2(u - x_1\alpha) \stackrel{?}{=} c_7 \quad (1)$$

If Eq. (1) is true, the received message is sent by the valid reader. Therefore, the authentication process is completed. Now, we divide the proposed scheme into two parts to show the correctness of the proposed scheme. The correctness is related to the tag side and the server side. The reader side is not important for the correctness since the burden of the reader is decreased and does not compute any values. Note that the reader generates a random value and transfers the values between the tag and the server. Therefore, the major authentication process is performed in the server side. The computations performed by the tag and the server are given in Fig. 2. In the server side, the server receives the values β, c_1, c_2 and c_3 from the tag and computes c_4, c_5, c_6 . In addition, the server checks c_2 whether it comes from the valid tag. Note that the essential calculations and authentication process are performed in the server side. As long as Eqs. (2), (3), (4) are satisfied, the proposed scheme is successfully run.

Correctness I: The server computes c_4 by using β which is sent by the tag. The correctness of this value is shown as follows:

$$\begin{aligned} c_4 &= AP_\sigma^{-1}\beta - y\alpha = AP_\sigma^{-1}(P_\sigma u + P_\sigma x\alpha) - y\alpha \\ &= A \cancel{P_\sigma^{-1}} \cancel{P_\sigma} u + A \cancel{P_\sigma^{-1}} \cancel{P_\sigma} x\alpha - y\alpha \\ &= Au + \underbrace{Ax}_y \alpha - y\alpha \\ &= Au \end{aligned} \quad (2)$$

Correctness II: With the following Eq. (3), the server checks the identification number of the tag. Note that the server generates c_4 in the previous state whereas c_3 is sent by the tag.

$$\begin{aligned} c_5 &= x' = c_3 - c_4 = Au + x - AP_\sigma^{-1}\beta + y\alpha \\ &= Au + x - Au \\ &= x \end{aligned} \quad (3)$$

Correctness III: To complete to the final controls, c_6 is computed by the server. Namely, the server makes sure with Correctness I and II where the server communicates the valid tag and the values are sent

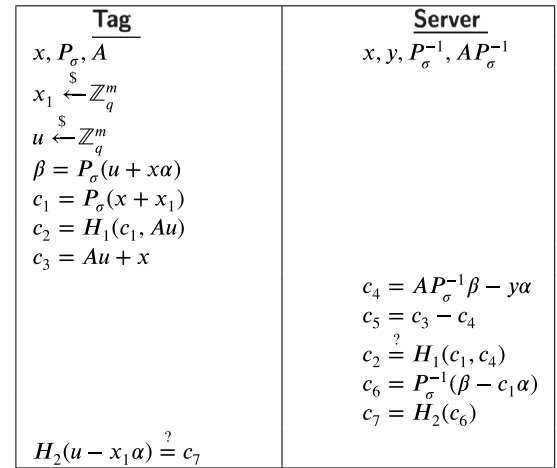


Fig. 2. The tag side and the server side.

by this valid tag. Thus, the server can compute c_6 by using the values β, c_1 and α as follows.

$$\begin{aligned} c_6 &= P_\sigma^{-1}(\beta - c_1\alpha) = P_\sigma^{-1}(P_\sigma(u + x\alpha) - P_\sigma(x + x_1)\alpha) \\ &= \cancel{P_\sigma^{-1}} \cancel{P_\sigma} u + \cancel{P_\sigma^{-1}} \cancel{P_\sigma} x\alpha - \cancel{P_\sigma^{-1}} \cancel{P_\sigma} x\alpha - \cancel{P_\sigma^{-1}} \cancel{P_\sigma} x_1\alpha \\ &= u + \cancel{x\alpha} - \cancel{x\alpha} - x_1\alpha \\ &= u - x_1\alpha \end{aligned} \quad (4)$$

3. Security analysis

In this section, we give a detailed the security analysis. We provide a post-quantum secure authentication scheme using the hardness of ISIS problem for IoT devices. Due to the fact that the identification scheme is the basis of authentication schemes, a lattice-based identification scheme given in [16] is used in the proposed scheme. Since the identification scheme in [16] is zero-knowledge, the proposed authentication scheme also satisfies zero-knowledge property. The tag and the reader can use the secret keys and ID. The transferred messages are not sent directly. Thus, anyone cannot get any useful information from the

transferred messages. Even if an adversary collects all messages, he/she is not able to extract the secret keys or ID. Owing to zero-knowledge property, the proposed scheme is completed only when the server verify the tag's identity. In addition, anyone cannot impersonate the tag or reader. For example, though an adversary can impersonate someone, the verification process is not completed and the communication will be rejected. For this reason, the proposed authentication scheme provides the following security goals.

- **Location Privacy:** Anyone cannot get information about the user's location. The proposed scheme has location privacy property discussed in [Theorem 1](#).
- **Unforgeability:** Unforgeability means that an attacker cannot impersonate any authenticator who gives the correct response to any challenges. Unforgeability property of the proposed scheme is shown in [Theorem 2](#).
- **Scalability:** Scalability is a desirable property for RFID schemes. The computational cost of the scheme should become constant or should not increase significantly even if the number of the tag is increased. Scalability property is proved in [Theorem 3](#) by considering the number of operations performed in the server side.
- **Forward Secrecy:** Forward secrecy means that any attacker cannot obtain previous information from all leaked transactions. Forward secrecy of the proposed scheme is given in [Theorem 4](#).
- **Resistance to Man In The Middle (MITM) Attack:** This is the type of attack in which the attacker eavesdrops the communication between the tag and the server and even changes the transferred messages. In [Theorem 5](#), the resistance to MITM attack is detailed.
- **Resistance to Replay Attack:** When an attacker stores all transferred messages, the attacker can replay the scheme with these previous messages. In [Theorem 6](#), it is shown that the proposed scheme is resistant to replay attack.
- **Resistance to Impersonation Attack:** In this attack, an attacker cannot impersonate the tag or the server. [Theorem 7](#) indicates that the scheme is resistant to impersonation attack.

Theorem 1. *The proposed scheme satisfies the location privacy property if and only if the commitments do not leak any information about the location.*

Proof. In the proposed scheme, the tag should generate the values β, c_1, c_2, c_3 to send response to the reader. Suppose that the tag always sends the same values in each session, it makes easy to trace the tag. In the proposed scheme, the tag generates random, different and independent values which are outputs of the cryptographically secure pseudo-random number generator and one-way hash functions for each session to prevent tracing. In addition, these values used in the proposed scheme are generated without the tag's location information. Therefore, any adversary cannot distinguish the transmitted message, which is sent by the same tag or not, and not afford to change the received message. Namely, the location is indistinguishable. This means that the proposed scheme is location independent. $\square$

Theorem 2. *If any attacker does not forge the tag's id number or change any value used in the scheme, the unforgeability property is satisfied.*

Proof. Note that the tag generates the following values to authenticate: $c_1 = P_\sigma(x+x_1)$, $c_2 = H_1(c_1, Au)$, $c_3 = Au+x$, $\beta = P_\sigma(u+x\alpha)$. We know that these commitment values are computed by using the tag's id number x , the values such as $u \xleftarrow{\$} \mathbb{Z}_q^m$, $x_1 \in \{0,1\}^m$, P_σ and A . Since the hash functions, the random value x_1 and the secret key u are used in the scheme, the attacker cannot get the truth values and he/she has no chance to fake any authenticator. Even if an attacker leak these values, the attacker cannot use the values in another session since the values are computed differently for each session. On the other hand, there exists no machine or attacker with non-negligible probability to solve

ISIS problem on which the proposed authentication scheme is based. Therefore, any attacker cannot extract the tag's id number from the leaked values. In addition, he/she will not be able to forge the tag's id number of the previous or future session by using the leaked values. This means that any attacker cannot forge the tag's id number and the values used in the scheme. The proposed scheme satisfies forward and backward unforgeability. $\square$

Theorem 3. *The performance or security of the proposed scheme does not depend on the number of tags and transactions. Then, the proposed scheme achieves the scalability.*

Proof. In the proposed scheme, the server stores each tag's id number and secret values in its database. These values are used directly by the server in the authentication phase. When the number of tags is large, or the size and traffic of the scheme are increased, it does not lead to the extra cost of the scheme. Therefore, the server has a constant burden. Namely, if new active members are added or the system is extended, the burden of the server will not change. The same computations are made for each tag. There is no extra computations for newly added tags. $\square$

Theorem 4. *Forward secrecy is achieved if and only if the session key is generated in each execution by considering the hardness of ISIS problem.*

Proof. We randomly choose session keys such as x_1 and α in scheme for each session. The secret key u is known only by the tag when A and y are shared keys between the tag and the server. Thus, anyone else or third party do not have any idea about the secret keys. Even if an attacker gets all transferred messages, he/she cannot reveal the secret keys. Solving the ISIS problem is only chance to get secret or session keys for the attacker who has all transferred messages. Note that the ISIS problem cannot be solved polynomial-time even in quantum computers [17]. Therefore, the attacker cannot obtain the secret or session keys. Assume that the attacker may obtain the secret keys with negligible probability, it does not guarantee that the previous secret keys can be found out with these revealed secret keys. Namely, the secret keys are updated after authentication. Therefore, the scheme satisfies forward secrecy. $\square$

Theorem 5. *A MITM attack is not possible under ISIS assumption by considering random value x_1 , tag id x and the tag's secret key u .*

Proof. Recall that the communication between the tag and the reader is insecure. On the other hand, the communication between the reader and the server is secure. Suppose that an attacker eavesdrops the communication, the attacker can get some values in that transaction. If the attacker wants to change the values and sends the tag/server, he/she cannot succeed. Note that the commitment messages are computed with the randomly generated $x_1 \in \{0,1\}^m$ or the secret key u for each session. In addition, each tag has unique ID number. With this respect, the authentication process will not be completed if the attacker generate any value instead of used one. The security of the proposed scheme depends on ISIS problem due to the unique identification number and the secret key. Therefore, the attacker's only chance to accomplish this attack is to solve the ISIS problem. We know that there is no algorithm to solve this problem in polynomial time even in quantum computers. Consequently, the proposed scheme has MITM resistance since the attacker cannot interfere with transferred messages by eavesdropping or changing messages. $\square$

Theorem 6. *A replay attack is not successful under the condition that α is randomly generated in each execution.*

Proof. An eavesdropper is able to collect all transferred messages between the tag and the reader in different protocols. It is impossible to

Table 2
The required number of (arithmetic) operations.

	Tag	Reader	Server
Chien and Huang [4]	1ROT+1XOR+1R+3H	1R	1ROT+1XOR+3H
Huang and Chung [8]	5XOR+4R+2CRC	1R	5XOR+2R+2CRC
Cho et al. [5]	1R+2H+2C+4XOR+4MOD	1R	2H+4C+6XOR+2MOD
Dass and Om [6]	1C+1XOR+ 2H+4R	3R	2R+1H+1XOR
Dong and Chen [7]	1R+4H+9C	1R+6H+17C+2MAC+2Enc	1R+1MAC+7C+1H
Zhen-yu Wu [9]	1R+2H+3M+1XOR	1R	1H+2M+1XOR
The proposed scheme	1R+2H+3M+2S	1R	1H+2M+2S

R: cost of pseudo-random generator, H: cost of hash functions, M: cost of matrix–vector multiplication, S: cost of vector–scalar multiplication, ROT: cost of rotation, CRC: cost of cyclic redundancy check, MOD: cost of the modular operation.

replay the scheme even the eavesdropper has all transferred messages. Considering the following states, the replay attack will fail.

- Assume that an eavesdropper sends the commitment values c_1, c_2, c_3, β which the eavesdropper get from the previous execution, to reader and pretends the tag. These values are transmitted the server by adding the random value α . Since the reader randomly generate α , the commitment values such as c_2 and c_3 do not match the in server side. Therefore, the eavesdropper cannot replay the scheme.
- In the second scenario, α obtained from the previous execution, is sent to the tag by the eavesdropper. Note that x_1 is randomly selected for each execution. The tag computes the commitment values with using the new selected x_1 and old α , and sends them to the reader. If the eavesdropper intervenes and sends the old α again, the replay attack is achieved. $\square$

Theorem 7. Impersonation attack is impossible since the hardness of the proposed scheme depends on the ISIS problem.

Proof. The impersonation attack can be considered in two cases such as the tag and the server.

- An attacker needs to compute c_1, c_2, c_3, β to impersonate the tag. However, the tag computes these commitment values with using $x_1 \in \{0,1\}^m$, x , P_σ and u . Note that these values are selected for each tag. Therefore, the attacker must solve ISIS problem to impersonate the tag. It is impossible to solve ISIS problem for the attacker. Then, the proposed scheme is resistant to tag impersonation attack.
- On the other hand, the attacker should eavesdrop the transferred messages and get the commitment values to impersonate the server. The attacker must reply the tag's commitment as the server. However, the server only knows the tag's secret values A, P_σ^{-1}, y and id number x . The attacker must solve the ISIS problem to succeed the impersonation attack. Therefore, the proposed scheme is resistant against the impersonation attack. $\square$

4. Parameter sets and comparison

In this section, we give parameter sets for 100-bit security level for the proposed authentication scheme and compare the proposed scheme with the previous ones in terms of efficiency and security properties. An attacker must solve the ISIS problem to attack the proposed scheme since the hardness of the proposed scheme is based on ISIS problem. Note that given A and y are public, the aim of the attacker is to find a vector x that satisfies $Ax = y \mod q$. The attacker also should find a solution vector u that ensures the hardness of ISIS problem (see Definition 1). For this reason, it is important to choose parameters. We use the set of parameters proposed in [16]. The parameters are chosen according to the complexity of attacks which find short lattice vectors in [16]. The chosen parameters are bounded the theorem proven in [18]. For a security level of 100-bit, m is 2048. In the proposed scheme, m -bit binary vectors are defined as private key and Hamming weight of the private key is the half of m . The other parameters are

Table 3
Security properties.

	[4]	[8]	[5]	[6]	[7]	[9]	Ours
Location Privacy	X	X	X	✓	✓	✓	✓
Unforgeability	✓	✓	✓	✓	X	✓	✓
Scalability	X	X	X	X	X	✓	✓
Forward Secrecy	✓	✓	✓	✓	✓	✓	✓
MITM Attack	✓	✓	✓	✓	✓	✓	✓
Replay Attack	✓	✓	✓	✓	✓	✓	✓
Impersonation Attack	✓	✓	✓	✓	✓	✓	✓
Quantum Resistant	X	X	X	X	X	X	✓

64, 257 and 256 for n , q and the output length of the hash function, respectively.

Remark 1. In the proposed scheme, the matrix–vector product has the most time-consuming arithmetic operation. Three matrix–vector product are performed in tag side. Therefore, the space requirement and time complexity are asymptotically in $O(n \times m)$, $O(n \times m)$, respectively. Note that the matrices are defined in $\mathbb{Z}_q$. The reader has no computation and only generates the random number in $\mathbb{Z}_q$ in constant time. Thus, the space requirement is $O(\log q)$. The server performs two matrix–vector product. The space requirement is $O(n \times m)$ in $\mathbb{Z}_q$ while the time complexity is $O(n \times m)$.

The efficiency of the authentication schemes is given in Table 2. Note that the communication costs of the addition and subtraction are negligible in Table 2. Note that the server computes the multiplication of AP_σ^{-1} offline and stores it. We significantly reduce the costs of the tag, reader and server when the proposed scheme compare [4–8]. The current scheme given in [9] is used pseudo-random number generator, hash function, matrix multiplication and XOR operator. In the proposed scheme, we use scalar multiplication instead of XOR operation. Therefore, the proposed scheme and [9] have almost same computation cost. Therefore, the proposed scheme is an alternative to Wu's authentication scheme [9].

Table 3 shows the comparison of schemes in terms of security properties. We give a detailed security analysis by comparing the schemes not only in terms of location privacy, unforgeability, scalability and forward secrecy but also MITM, replay, impersonation and quantum attacks. The schemes given in [4–9] are not secure against quantum attack whereas the proposed scheme satisfies all security requirements needed for RFID systems.

5. Conclusion and future works

We propose an authentication scheme for IoT devices. We give the correctness of the proposed scheme. In addition, we present detailed security analysis for some attacks such as man in the middle, impersonation and replay attacks. Consequently, the proposed scheme is compared with the previous ones in terms of computation costs in each part, tag, reader and server, respectively. The proposed scheme is based on ISIS problem. Therefore, the scheme is resistant to post-quantum attack and can be securely implemented RFID application

for post-quantum. Hardware implementation is left as a future work. Moreover, complexity of the arithmetic operations can be improved for the target platform for the proposed authentication scheme.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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